

132 Pattern Problem - Explanation

Problem: Find if there exist $i < j < k$ such that $nums[i] < nums[k] < nums[j]$. Approach: We use a stack and traverse from right to left. Key Idea: - stack stores possible $nums[j]$ - second stores $nums[k]$ Code:

```
class Solution:
    def find132pattern(self, nums):
        stack = []
        second = float('-inf')

        for i in range(len(nums) - 1, -1, -1):
            if nums[i] < second:
                return True

            while stack and nums[i] > stack[-1]:
                second = stack.pop()

            stack.append(nums[i])

        return False
```

Explanation: - Traverse from right to left - If current number is less than 'second', pattern found - Pop from stack to update 'second' - Push current element to stack Time Complexity: $O(n)$ Space Complexity: $O(n)$